

Question 4

Suppose that we use some statistical learning method to make a prediction for the response Y for a particular value of the predictor X . Carefully describe how we might estimate the standard deviation of our prediction.

Answer

We can use the bootstrap to estimate the standard deviation of our prediction. The idea is to repeatedly generate new datasets from our original training data, refit the model each time, and see how much the prediction for our particular value of X changes across those different fits.

More specifically, we resample our original dataset with replacement to create a bootstrap dataset of the same size as the original (some observations will show up more than once, others might not appear at all). We fit our statistical learning method on this bootstrap dataset and use it to predict Y at our value of X . We repeat this process a large number of times, say B times, which gives us B different predictions $\hat{Y}_1, \hat{Y}_2, \dots, \hat{Y}_B$ for the same X .

Since each bootstrap sample is slightly different, these B predictions will vary a bit around the true prediction. We estimate the standard deviation of the prediction using

$$\text{SE}(\hat{Y}) = \sqrt{\frac{1}{B-1} \sum_{b=1}^B (\hat{Y}_b - \bar{Y})^2}$$

where $\bar{Y} = \frac{1}{B} \sum_{b=1}^B \hat{Y}_b$ is the average of the B bootstrap predictions.

This works because it mimics what would happen if we had access to many different training sets drawn from the population, something we don't actually have in practice since we usually only get one dataset. The bootstrap lets us approximate that variability using just the one dataset we do have.